9.2 Combined seals

Several seal types are used in combination for a seal application under an environment where dust, water content, etc. exist so much or for mechanical portions which are not allowed to be contaminated by lubricant leak.

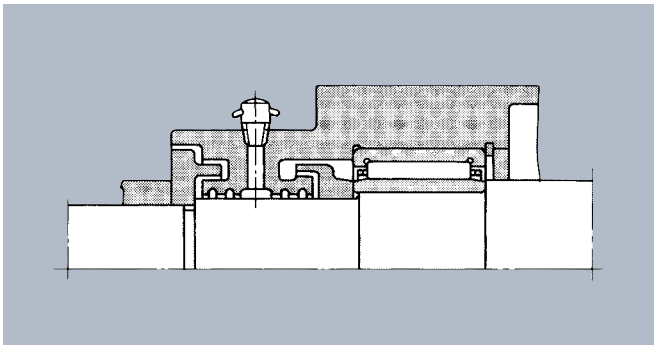


Fig. 9.11 Combined non-contact seal
Combination of labyrinth seal and oil groove seal

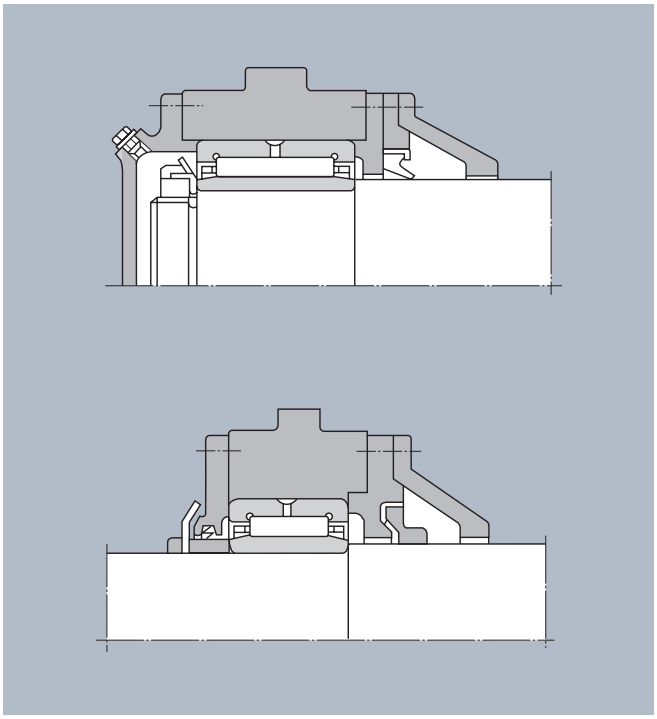


Fig. 9.12 Combined seal
Combination of contact seal and non-contact seal

9.3 Clearance setting

Oil groove seal and labyrinth seal have the better sealing effect as the shaft - housing clearance gets the smaller, but the actual clearance is generally selected from the following clearance values, under consideration of machining and assembling conditions, shaft deformation, etc.

Table 9.3 Clearances

Seal type	Shaft diameter	Radial clearance	Axial clearance
Oil groove seal	50 and less	0.2～0.4	———
	Over 50 to 200	0.5～1.0	
Labyrinth seal	50 and less	0.2～0.4	1.0～2.0
	Over 50 to 200	0.5～1.0	3.0～5.0

9.4 NTN seals

Special-purposed **NTN** seals are available for the needle roller bearings.(Refer to **Table 9.4** on page A-48.) For the more detailed information refer to "Dimensions Table" on page B-233.

9.5 Seal materials and corresponding operating temperature ranges

The oil seal lip is ordinarily made of nitril rubber, but acrylic rubber, silicone rubber and fluoro-rubber are used as the lip material according to atmosphere temperature, an objective to be sealed, etc. **Table 9.5** shows the operating temperature ranges available for the respective materials.

Table 9.5 Seal materials and corresponding operating temperature ranges (Reference)

Seal materials	Operating temperature ranges ˚C
Nitril rubber	−25～+100
Acrylic rubber	−15～+130
Silicone rubber	−70～+150
Fluoro-rubber	−30～+180
Felt	−40～+120

9.6 Seal types and allowable speed

The allowable speed for the contact seal type depends on the surface roughness, accuracy and lubrication property of sliding surface, operating temperature, etc. **Table 9.6** shows the allowable speed every each seal type, as a guideline.

Table 9.6 Seal types and corresponding allowable speed (Reference)

Seal types	Allowable peripheral speed m/s
Felt seal	4
Grease seal	6
Oil seal (Nitril rubber)	15
Oil seal (Fluoro-rubber)	32
V-ring seal	40

Table 9.4 Seals (NTN contact seals)

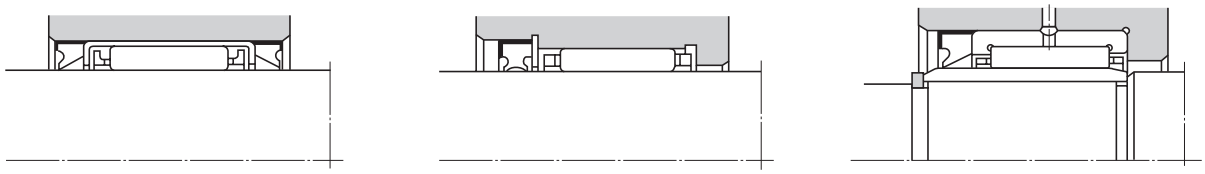
Seal type	**Contact seals (G type, GD type)**
	Seal using mainly direct contact
Features	This seal type is a special-purposed seal for needle roller bearings which was designed for smaller section height so as to correspond flexibly to the said bearings. This is a synthetic rubber contact seal reinforced with steel plate, being then used in the operating temperature range of -25 to +120˚C and, under continuous running condition, used at 100˚C and less. For application under special operating conditions of over 120˚C at operating temperature, etc., feel free to contact **NTN** for any inquiry.
Application examples	

Fig. 9.13 Bearing sealing by NTN seals (Example)

9.7 Shaft surface roughness

Sealing performance and seal life depend on the surface roughness, accuracy and hardness of shaft sliding surface with which the seal lip comes in contact. **Table 9.7** shows the surface roughness as a guideline. For improving better the wear resistance of shaft surface it is desirable to secure the shaft surface hardness at least at HRC40 and over (at HRC55 and over if possible) by heat treatment and hard chrome plating.

Table 9.7 Shaft surface hardness (Reference)

Peripheral speed m/s		Surface roughness
over	incl.	Ra
	5	0.8a
5	10	0.4a
10		0.2a

10. Bearing Handling

A rolling bearing is a precision part. It must be handled very carefully to keep its preciseness unchanged. Particularly the handling instructions given hereunder must be observed in handling.

[1] Keeping bearing and other related parts clean
Foreign matters such as mist, dust, etc. would cause harmful affect on bearing running and life. To avoid such harmful effect, bearing and other parts mounted adjacent to the bearing must be kept clean and, in addition, handling tools, lubricant, washing oil, work environment, etc. must always be maintained in clean condition respectively.

[2] Careful handling
Any shock to a bearing in handling would result in surface flaw and indent of its raceway surface and rolling elements and, in the worst case, result in cracking and chipping. To avoid such defects and trouble, bearings must be handled with good care.

[3] Use of proper handling tools
It should be avoided to use another inappropriate tool as an alternative of the specific tool in installing and removing.
Specific tools suited to the individual bearing types must be used. The special-purposed handling tools must be used particularly when installing a drawn cup needle roller bearing.

[4] Protection of bearing from rusting
A rust preventive oil is coated on all of bearings, as a rule. However, direct handling of bearings by the bare hands would cause rusting of the bearings from adhesion of hand sweat thereto. To protect bearings from rusting, a pair of gloves must be put on or otherwise mineral oil be coated on the hands in the case of direct handling by the bare hands.

10.1 Bearing storage

Bearings are all shipped after coated with rust inhibitor and packaged. It is desirable to store bearings at room temperature and relative humidity of 60% and less. The storage period of a grease-filled and sealed bearing is limited to the specific lifetime of non-greased bearing.

10.2 Installation

Although depending on bearing type and fitting conditions, the bearing installing methods described hereunder are selectively available as the general method. However, refer to Commentary given in the Dimensions Table for installation of the drawn cup needle roller bearings.

(1) Preparations prior to installing
For installation of bearings it is desirable to prepare a cleaned and dried work place.
Contaminant, burr, chips, etc. must be removed completely from all the parts related to a shaft and a housing before installing.
Furthermore, the bearing mount must be inspected to verify whether its dimensional accuracy, profile accuracy and surface roughness are within the respective specified tolerance ranges.
Bearings are all unpacked before installing. In the case of using bearings with grease lubrication, generally they may be installed as are without washing off the rust inhibitor coated thereon.
Where a bearing is used with oil lubrication or the lubricating function of a grease-lubricated bearing is impaired by a mixture of grease and rust inhibitor, however, the bearing must be installed after complete removal of the rust inhibitor by clean washing oil.
It is not allowed to wash shield type and seal type bearings and one-way clutch.

(2) Press-fit by a press machine
In general, the press-fit method using a press machine is widely used for installation of bearings. In detail, the bearing ring (inner ring or outer ring) is press-fitted slowly via a backing strip as illustrated in **Fig. 10.1**. **It is not allowed to apply the press force to a bearing via its rolling elements like a bad example illustrated in Fig. 10.2 (page A-50)**.
Further, a small size bearing with small interference may be installed by hammering the bearing ring with a plastic hammer or the like. **In that case, however, the uniform hammering force must be applied to the bearing side face via the backing strip as illustrated in Fig. 10.1, because direct hammering to the bearing end face or partial hammering by use of a punch could impair the specific bearing performance.**
Fitting the inner ring on a shaft by hammering the outer ring or fitting the inner ring by hammering itself would result in surface flaw and indent of the raceway surface and the rolling elements. Therefore, doing so must be avoided absolutely.
Also, friction of the fitting surface can be reduced by coating high viscosity oil on the surface.

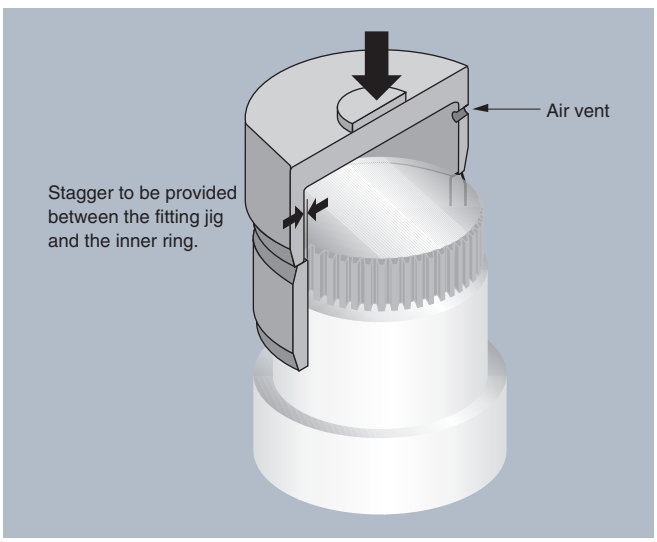


Fig. 10.1 Press-fitting of inner ring

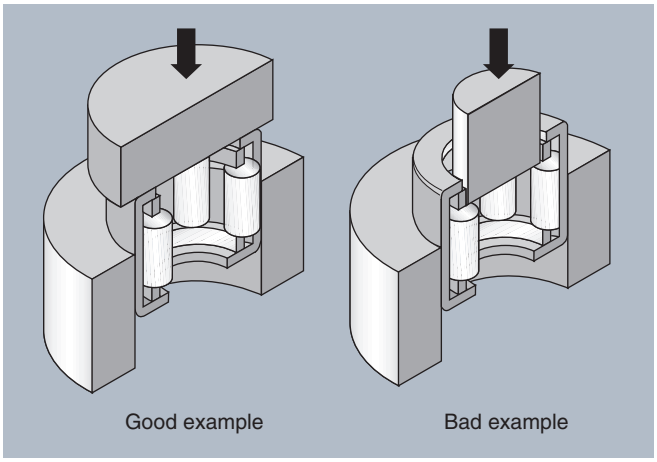


Fig.10.2 Press-fit

(3) Shrinkage fit

This shrinkage fit method is to fit the inner ring on a shaft after dipping it in hot oil for thermal expansion of its bore. This method is also widely used for installation of bearings. In heating the inner ring, low-corrosive mineral oil or the like is used. In this case, the heated inner ring is natural-cooled down after shrinkage-fit, but simultaneously it shrinks in axial direction, too. Therefore, the inner ring must be pressed to the shaft shoulder until adequate cool-down, so as not to allow a clearance between the inner ring and the shoulder.

In any case, however, it is not allowed to heat the inner ring up to over 120˚C.

Fig. 10.3 shows the relationship of inner ring bore thermal expansion to heating temperature.

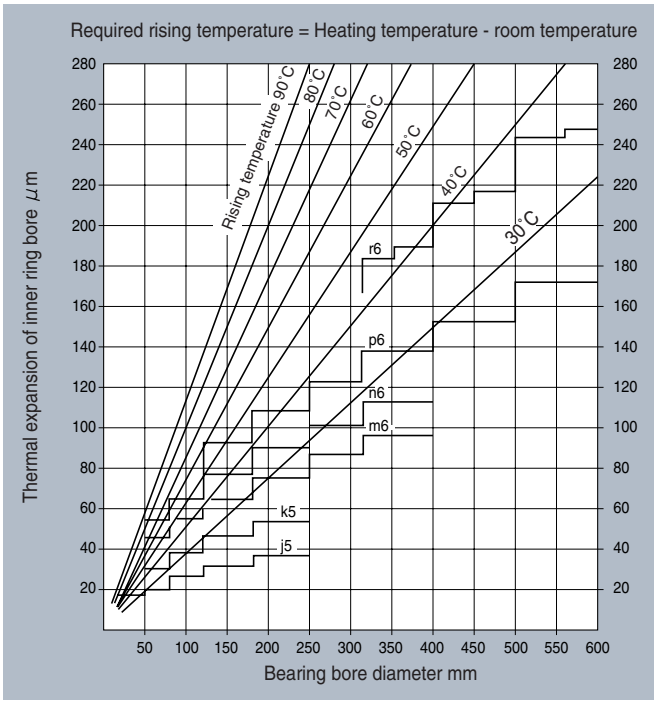


Fig. 10.3 Rising temperature required for shrinkage fit of inner ring

10.3 Post-installation running test

Bearing running test must be conducted on an installed bearing to check whether it was installed correctly. In this case, accelerating immediately the running speed up to the rated speed must be avoided absolutely. Doing so could result in damage of the bearing in the case of imperfect installation and seizure of the bearing in the case of inadequate lubrication. In testing, therefore, the shaft or the housing must first be rotated by hand and thereafter the shaft be rotated at low speed under no-load condition by the driving power, unless any failure is detected upon checking. After that, the running speed and load on the bearing must be accelerated and increased gradually while checking the running condition.

Running sound level and tone of a bearing can be checked by a sound scope held in contact with the bearing housing. The sound is normal if it is a pure sound upon checking. In this case, high metallic sound or irregular sound from the bearing, if any, reveals occurrence of somewhat failure. In such a case, possible cause of the failure can be presumed by measuring quantitatively vibration amplitude and frequency using a vibrometer.

Generally bearing temperature is presumed from the circumferential temperature of a housing. However, it can be judged more exactly if possible to measure directly the outer ring temperature by utilizing the oil hole thereof, etc.

Needless to say, bearing temperature rises with elapse of the running time and reaches a constant level after a certain time. Rapid bearing temperature rise or continued temperature rise in excess to the constant level or abnormally high bearing temperature would reveals occurrence of somewhat failure. In such a case, careful check is required.

Table 10.1 shows the required check items.

Table 10.1

Hand operation	Variation in torque Over-torque Chattering Abnormal sound	Imperfect installation Under-clearance, great seal friction, etc. Indent and flaw on raceway surface Inclusion of dust and other foreign matter
Power operation	Abnormal noise and vibration Abnormal temperature	Inclusion of dust and other foreign matter, indent on raceway surface,over-clearance, inadequate lubrication Use of improper lubricant, imperfect installation, under-clearance

10.4 Bearing removal (dismounting)

Some related bearings are removed from each shaft, incurred by periodic disassembly or incidental trouble of a machine. In this case, these bearings must be handled with good care similarly to the case of installation, when they are reused or the condition thereof is examined after removed (dismounted). Too tight fit of a bearing would incur difficulty in dismounting and, therefore, the bearing-related construction must be fully considered in design of

the bearing fixing method.

Regarding the dismounting method, generally the press method (**Fig. 10.4**) and the puller method (**Fig.10.5**) are available for dismounting inner ring though depending on bearing type and fitting conditions.

When inner ring is dismounted from a shaft by heating, good care must be exercised not to over-heat the inner ring.

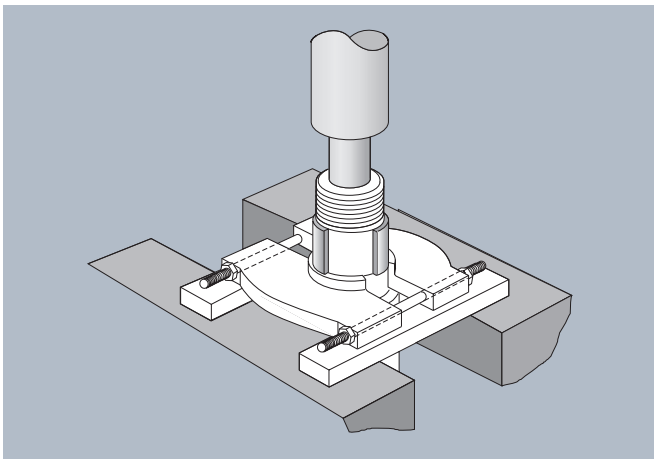


Fig. 10.4 Bearing removal by a press machine

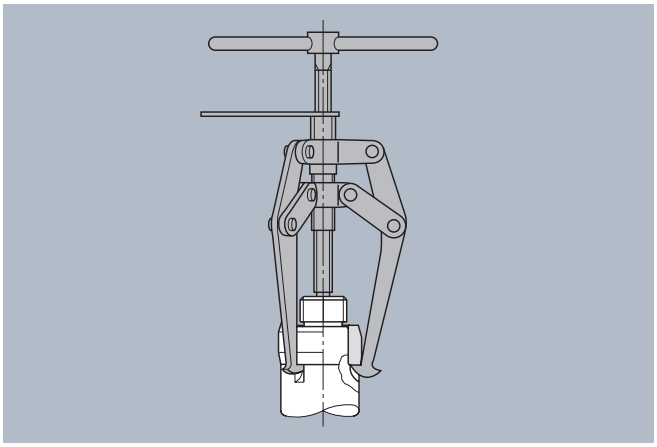


Fig. 10.5 Bearing removal by a puller

10.5 Force required for press-fit and pull-out

The force required for press-fitting or pulling out inner ring on / from a shaft can be determined by **formula (10.1)**.

$$K_a = f_K f_E \frac{d}{d+3} \Delta d_F \cdots\cdots (10.1)$$

Where,

K_a : Force required for press-fitting or pulling-out N (kgf)

f_K : Resistance factor being determined by shaft to inner ring friction factor

For press-fitting····· 40 (4)

For pulling-out·····60 (6)

f_E : Coefficient depending on inner ring dimension

$$f_E = B\left[1-\left(\frac{d}{F_1}\right)^2\right]$$

B : Inner ring width mm

d : Inner ring bore diameter mm

F_1 : Mean outer diameter of inner ring mm

Δd_F : Apparent interference μm

Actual press-fit force and pull-out force could eventually exceed the respective calculate value due to installing error. Hence, it is recommended to design the dismounting tools so as to have the strength (rigidity) resistible to a load 5 times as much as the calculated press-fit force and pull-out force.

10.6 Washing

Rotating a bearing with foreign matter adhered thereto could result in damage of the raceway surface, etc. Therefore, any dismounted bearing is usually washed by light oil or kerosene or the like for complete removal of foreign matter. In this case, two washing vessels must be used as follows; one for rough washing and another for finish washing. Rough washing is done for removal of oil and foreign matter from bearing, while finish washing is done for fine washing of the roughly-washed bearing. Further, any vessels used for washing must be provided with a steel net as the middle bottom above the vessel bottom, as illustrated in **Fig. 10.6**, to prevent the bearing from coming in direct contact with the vessel bottom.

Furthermore, rust preventive treatment must be applied to the washed bearing immediately after washing, to thereby protect it from a risk of corrosion.

In addition, the legal requirements as applicable (environmental preservation, industrial labor safety and health, etc.) and the washing specification submitted from a detergent manufacturer must be taken in consideration in washing.

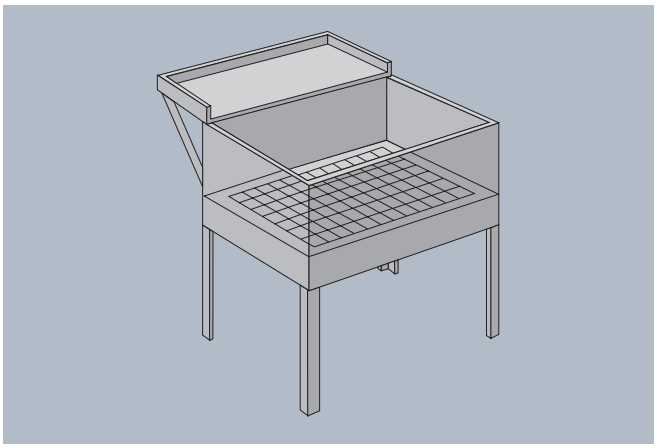


Fig. 10.6 Washing tank

11. Technical Data

11. 1 HL Bearing

The form of bearing separation life can be mainly classified into the internal origin type and the surface origin type. Bearing separation of the surface origin type is said to occur in an inadequately lubricated area and it is widely recognized that the bearing separation life correlates to the oil film parameter that can be determined by "Elastic Fluid Lubrication Theory (EHL Theory)".

This oil film parameter must be increased for controlling the surface origin type separation and, for that, the bearing manufacturers have made possible effort for better improvement of lubricant and for upgrading of the bearing rolling raceway surface roughness.

In recent years it has been proposed by the relative field that the capability of forming an oil film on contact surfaces could be improved better by changing the form and directivity of surface roughness of machined parts.

On the other hand, **NTN** developed successfully long life HL (High Lubrication) bearings under "Micro EHL Theory" as one of the countermeasures against surface origin type separation. And these HL bearings have been widely used in each field since developed.

11.1.1 Basic concept of HL bearing

Fig. 11.1 illustrates the basic concept of the HL bearing.

In this Figure, the hatched area shows a contact portion deformed elastically and the dotted line shows the lubrication flow, based on "Internal lubrication fluid flow model in contact portion" disclosed by H.S. Cheng and others.

Lubricant flow in (B) has a resistance greater than that in (A). This means that the quantity of a fluid existing internally in the contact portion increases. In other words, the thickness of an oil film formed on the rolling contact surface increases with increase of the fluid quantity.

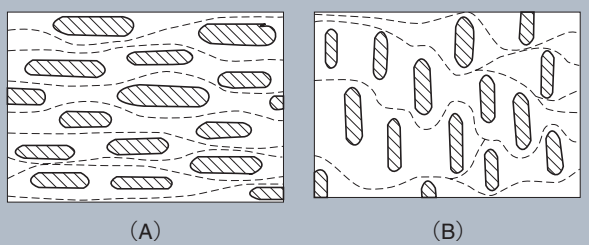


Fig. 11.1 Directivity of finished surface and lubrication fluid flow model

11.1.2 HL surface

Fig. 11.2 is a magnified photo showing roller surface. In this Figure the blackened spots are micro recessed spots and, as seen from this Figure, a number of micro oil pots of around several tens mm exist at random. The recessed spots on this surface can be modified into any optional size and the number of spots by changing the machining conditions. **Fig. 11.3** shows the bus line form of the HL surface, from which the depth of the micro recessed spots is nearly 1 mm.

11.1.3 HL bearing application examples

The HL surface-treated bearings are widely used in various fields. For example, they are used for car transmission, hydraulic devices, various reduction gears, etc.

As an example of special application, HL surface treatment is applied to the special-purposed bearing for the rocker arm of car engine. This is highly appreciated as an effective seizure preventive measure.

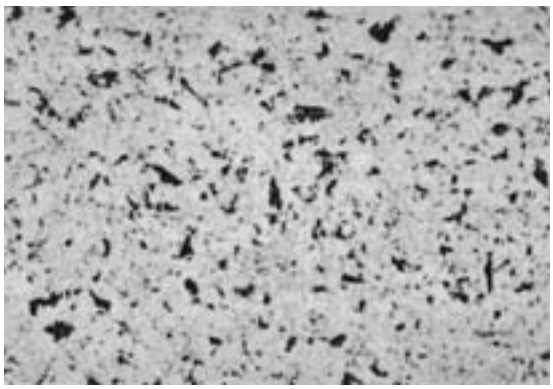


Fig. 11.2 Magnified photo showing roller surface

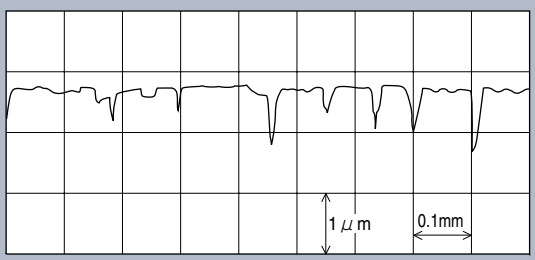


Fig. 11.3 Bus line form of HL surface

11.2 Bearings with Solid Grease

Bearings with Solid Grease means a bearing in which thermal solidification type grease (mainly composed of lubrication grease and resin) is filled as a lubricant material.

Once this thermal solidification type grease (referred to as Bearings with Solid Grease) was heated and cooled down (hereinafter referred to "heat treatment"), the paste-like grease hardens as much lubricant held unchanged.

The Bearings with Solid Grease is solidified after heat treatment so that the lubricant does not leak easily even in the case wide amplitude vibration and strong centrifuge act on the bearing, thus contributing to prevention of lubricant leak and longer bearing life.

The spot pack specification in which Bearings with Solid Grease was packed at multiple points on the cage and the full pack specification in which Bearings with Solid Grease was nearly full-packed in the spacing volume of bearing are both available for the Bearings with Solid Grease. Further, the needle roller bearings are all based on the full pack specification.

11.2.1 Characteristics of Bearings with Solid Grease

(1) Less lubricant leak

Bearings with Solid Grease holds much lubricant internally in bearing because it is solidified after heat treatment. This lubricant is gradually fed onto the rolling surface by heat generating from the bearing and centrifuge, whereby lubricant leak is minimized. This serves to prevent the peripheral environment from being contaminated, comparing with general lubrication greases.

(2) Good lubricating characteristic

Even when wide amplitude vibration and strong centrifuge act on a bearing, lubricant does not leak easily and, in addition, Bearings with Solid Grease does not emulsify and flow out easily, because of its solid type, even in the case of water content inflow to bearing.

Thus, the lubricating characteristic is more excellent than general lubrication greases.

(3) Sealing effect

Bearings with Solid Grease acts as a preventive barrier against foreign matters (water content, dust, etc.) invading from outside, but does not function perfectly as a seal. Therefore, use of the contact type rubber seal (for deep groove ball bearing, bearing unit) or another seal (for other bearings) is recommended particularly when high sealing performance is required.

11.2.2 Attentive points in using Bearings with Solid Grease

(1) The needle roller bearing dimension to accept Bearings with Solid Grease differs every each bearing type. Feel free to contact **NTN** for the detail.

(2) The allowable temperature range is -20˚C to 80˚C, but 60˚C and less for continuous long term running. Fully note the operating temperature.

(3) Allowable rotational speed $F_w \cdot n$ value (F_w = inscribed circle diameter [mm]) $\times$ (n= operating rotational speed [r/min]) is differs from that in use of general grease and lubrication oil.
Fully note this matter.
Use any needle roller bearings on $F_w \cdot n$ value = 30,000 and less.

(4) For the bearings of the full pack specification, minimum load equivalent to at least around 1% of the basic dynamic load rating is needed for enabling the rolling elements to run without slipping.

(5) Bearings with Solid Grease acts as a preventive barrier against foreign matter invading from outside, but does not function perfectly as a seal. Therefore, combined use of Bearings with Solid Grease and rubber seal is recommended particularly when high sealing performance is required.

11.2.3 Application examples of Bearings with Solid Grease bearings

- Bearing for the paper feeder of a printing machine
- Bearing for the mast roller guide of a forklift
- Support bearing for the swing arm of a two-wheeled car
- Bearing for a machine tool
- Guide bearing for the guide unit of a press machine
- Bearing for the link mechanism of an automatic loom
- Bearing for the conveyor guide of a food packaging machine



11.3 Calculation Examples

11.3.1 Shrinkage factor and post-installation clearance of drawn cup needle roller bearing

The recommended fit data for the standard bearings is as described in **Table 1** on page B-33. This paragraph describes hereunder the calculation methods to be used when the bearing fit conditions are reviewed in detail.

1) Calculation of bearing shrinkage factor

For the drawn cup bearings, the shrinkage factor is calculated using the following method.

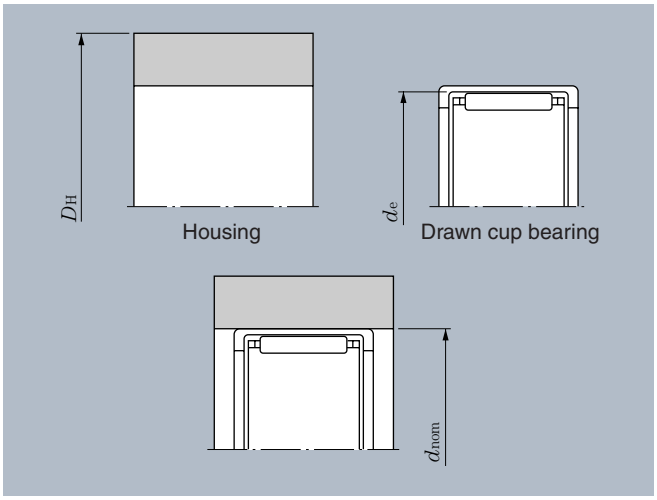


Fig. 11.4

$$\lambda = \frac{2t}{E_2} \cdot \frac{1-S^2}{\dfrac{(0.7S^2+1.3)(1-t^2)}{E_1} + \dfrac{(0.7+1.3t^2)(1-S^2)}{E_2}} \quad \cdots\textbf{(11.1)}$$

Where,

λ : Outer ring shrinkage factor
D_H : Housing outer diameter mm
d_{nom} : Nominal diameter of fitting portion mm
d_e : Rolling surface diameter of outer ring mm
E_1 : Modulus of housing vertical elasticity (Young's modulus) MPa (kgf／mm²)
E_2 : Modulus of outer ring vertical elasticity (Young's modulus) 2.07×10^6MPa (21 200kgf／mm²)

$$S = \frac{d_{nom}}{D_H}$$

$$t = \frac{d_e}{d_{nom}}$$

2) Inscribed circle diameter after complete bearing fit in the housing on actual machine

[1] Inscribed circle diameter in press-fitting of master ring

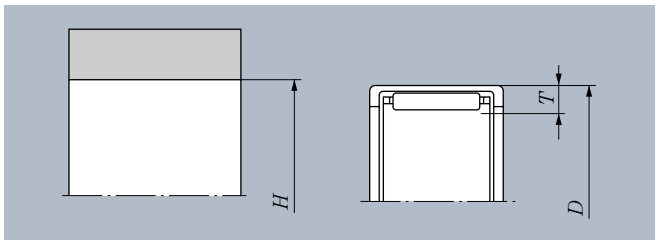


Fig. 11.5

H : Housing inner diameter mm
T : Roller diameter + plate thickness mm
D : Outer diameter of drawn cup needle roller bearing mm
L_i : Post press-fit inscribed circle diameter mm

When the master ring is press-fitted, the dimension of "roller diameter + plate thickness" remains unchanged. Hence, the inscribed circle diameter L_i is determined by the following **formula**.

$$L_i = D-2T-\lambda(D-H) = (1-\lambda)D-2T+\lambda H \quad \cdots\textbf{(11.2)}$$

Determine the mean value of "roller diameter + plate thickness" (=T) and standard deviation from **formula (11.2)**. The mean value of **formula (11.2)** is determined as follows.

$$m_{Li} = (1-\lambda)m_D - m_{2T} + \lambda m_H \quad \cdots\textbf{(11.3)}$$

Standard deviation of formula **(11.2)**

$$\sigma_{Li2} = (1-\lambda)_2 \cdot \sigma_{D2} + \sigma_{2T2} + \lambda_2 \sigma_{H2} \quad \cdots\textbf{(11.4)}$$

In the case of master ring, due to $\sigma_{H2}=0$ the **formula (11.4)** is expressed as follows.

$$\sigma_{Li2} = (1-\lambda)_2 \cdot \sigma_{D2} + \sigma_{2T2} \quad \cdots\textbf{(11.5)}$$

The unknown values in **formulas (11.3)**, **(11.5)** are only m_{2T} and σ_{2T}^2. Hence, substitute the known numerical values for **formulas (11.3)**, **(11.5)** to determine m_{2T} and σ_{2T}^2.

[2] Even when bearing ring is press-fitted in the housing on actual machine, consider the inscribed circle diameter similarly to the master ring press-fit. Herein, the calculation formulas for press-fit in the housing on actual machine can be discriminated as follows from **formula (11.3)**, **(11.4)** by adding " ' " to each formula.

$$m_{Li}' = (1-\lambda')m_D - m_{2T} + \lambda' m_H' \quad \cdots\textbf{(11.6)}$$
$$\sigma_{Li}'^2 = (1-\lambda')_2 \cdot \sigma_{D2} + \sigma_{2T2} + \lambda'^2 \sigma_{H}'^2 \quad \cdots\textbf{(11.7)}$$

[3] For m_{2T} and σ_{2T}^2 in **formula (11.6)**, **(11.7)**, substitute the values determined previously for the respective formula.

[4] From the calculations, the inscribed circle diameter in press-fitting in the housing on actual machine can be

expressed in the following **formula**

$$L_i' = m_{Li}' \pm 3\sigma_{Li}' \quad \cdots\cdots (11.8)$$

[5] Radial internal clearance can be determined considering the mean value and standard deviation of shaft in **formulas (11.6)**, **(11.7)**.

[6] The aiming radial internal clearance value is generally set up so an ordinary clearance can be got. However, the recommended clearance values are available every the individual portions in the case of bearing application to automobile. Feel free to contact **NTN** for the detail.

11.3.2 Track load capacity of cam follower and roller follower

The reference hardness (reference tensile stress) was set up from the relationship between the follower hardness and net tensile stress of the material and the track load capacity was determined

from the relationship of the setup reference stress to hertz stress.

How to set up the reference hardness (tensile stress) differs a little bit depending on each bearing manufacturer. Herein, the relevant Table appended to "JIS Handbook for Irons and Steels" was used as the hardness - tensile stress relationship.

(Approximate numerical value under JIS Z8413 Revised Conversion Table)

For HRC40, σ = 1.245MPa (127kgf/mm^2) was adopted as the reference hardness (tensile stress).

<Track load capacity adjustment factor>

The tensile stress strength of the follower material increases with increase of its hardness, and the track load capacity of the follower increases incurred by increase of the tensile strength strength. In this case, the load capacity increase rate can be determined by multiplying the track load capacity by applicable track load capacity adjustment factor shown in **Table 11.1**.

Note) The track load capacity determined herein is based on net tensile stress as the reference, not allowable hertz stress. Generally stress (specific stress) resulting in creep of follower material is greater than the tensile stress. Particularly in the case of static load, this track load capacity comes to a safety side value.

[Ex.] Determination of load capacity T_c' of track with certain hardness by use of track load capacity adjustment factor.

Assuming track load capacity described in Dimensions Table as T_c and track load capacity adjustment factor at applicable hardness as G respectively, the track load capacity T_c' can be determined as follows.

$$T_c' = G \cdot T_c$$

For hardness HRC50 at NATR15X,

T_c = 11 900N (1 220kgf), G = 1.987

$\therefore T_c'$ = 1.987 × 11 900N (1 220kgf)

= 23 645N (2 424kgf)

Reference (Track load capacity calculation process)

- For a cylindrical outer ring

$$\sigma_{max} = \frac{1}{60.9}\sqrt{\frac{T_c \Sigma\rho}{B_{eff}}}$$

- For spherical R outer

$$\sigma_{max} = \frac{187}{\mu_\nu}\sqrt[3]{(\Sigma\rho)^2 T_c}$$

σ_{max} = 1 245MPa (127kgf/mm^2)

T_c : Track load capacity N (kgf)

$\Sigma\rho$: Sum of curvature

B_{eff} : Effective contact length mm

Herein (Outer ring width - chamfer)

μ_ν : Factor being determined by curvature

Table 11.1 Track load capacity adjustment factor

Hardness HRC	Tensile strength MPa {kgf/mm^2}	Adjustment factor G	
		for cylindrical outer ring	for spherical R outer ring
20	755 {77}	0.368	0.223
21	774 {79}	0.387	0.241
22	784 {80}	0.397	0.250
23	804 {82}	0.417	0.269
24	823 {84}	0.437	0.289
25	843 {86}	0.459	0.311
26	862 {88}	0.480	0.333
27	882 {90}	0.502	0.356
28	911 {93}	0.536	0.393
29	931 {95}	0.560	0.419
30	951 {97}	0.583	0.446
31	980 {100}	0.620	0.488
32	1 000 {102}	0.645	0.518
33	1 029 {105}	0.684	0.565
34	1 058 {108}	0.723	0.615
35	1 078 {110}	0.750	0.650
36	1 117 {114}	0.806	0.723
37	1 156 {118}	0.863	0.802
38	1 176 {120}	0.893	0.844
39	1 215 {124}	0.953	0.931
40	1 245 {127}	1.0	1.0
41	1 294 {132}	1.080	1.123
42	1 333 {136}	1.147	1.228
43	1 382 {141}	1.233	1.369
44	1 431 {146}	1.322	1.519
45	1 480 {151}	1.414	1.681
46	1 529 {156}	1.509	1.853
47	1 578 {161}	1.607	2.037
48	1 637 {167}	1.729	2.274
49	1 686 {172}	1.834	2.484
50	1 754 {179}	1.987	2.800
51	1 823 {186}	2.145	3.141
52	1 882 {192}	2.286	3.455
53	1 950 {199}	2.455	3.847
54	2 009 {205}	2.606	4.206
55	2 078 {212}	2.787	4.652

11.3.3 Outer ring strength

Generally any outer ring never breaks down as long as the load acting it is a usual operating load. This paragraph describes hereunder the strength calculation method to be used when the outer ring strength under shock load and heavy load is reviewed.

The outer ring strength can be determined using the formula given hereunder, assuming the respective outer ring profiles as illustrated in **Fig. 11.6**. In this case, the outer ring rupture strength means the bridged rupture strength of roller.

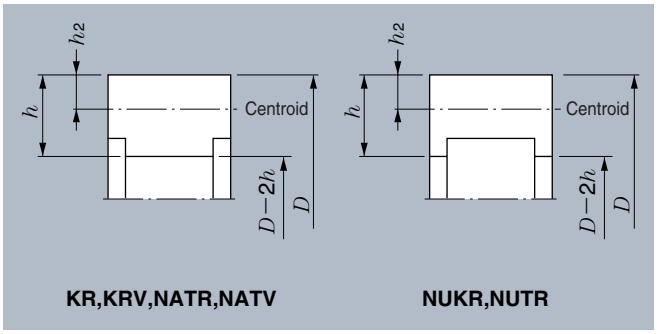


Fig. 11.6

Regarding how to set up breaking stress, in general 1760MPa (180kg/mm²) can be set up as the breaking stress for bearing steel, but it is desirable to set up the breaking strength with safety-side value (1170MPa [120kgf/mm²]), where stress concentration is taken into account. Generally any outer ring never break down as long as the load acting on it is usual operating load, but it necessary to check the rupture structure of outer ring, where shock load and heavy load act on it.

$$P=\frac{4\pi}{1+f(\alpha)}\times\frac{D-2h}{h\,(D-2h^2)^2}\times I\times\sigma$$

Where,

$$f(\alpha)=\frac{(\pi-\alpha)\sin\alpha-(1+\cos\alpha)}{2\cos\alpha}$$

$$\alpha=\frac{\pi}{Z}\ (\text{rad.})$$

P : Breaking load (N)
I : Secondary moment of outer ring section (mm⁴)
Z : Number of rollers
σ = Breaking stress (MPa)
D, h, h_2 : per **Fig. 11.6** (mm)

11.3.4 Stud strength of cam follower

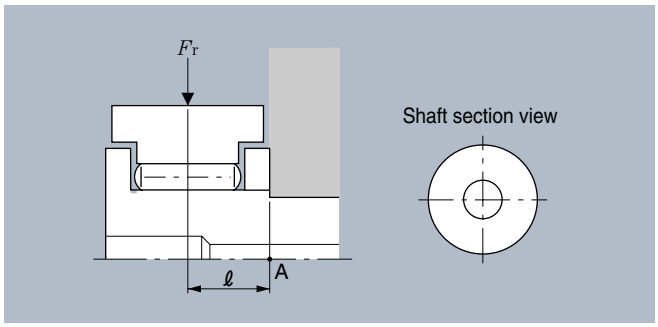


Fig. 11.7

When load Fr acts on the center point of outer ring as illustrated in **Fig.11.7**, bending moment $F_r \cdot \ell$ generates and consequently bending stress σ_1 (deemed as tensile stress) acts on the stud surface. In addition to this bending stress, tensile stress σ_2 generates from screw tightening because the stud itself is clamped to machine body with nuts. The stud strength can be reviewed from comparison of the sum ($\sigma_1 + \sigma_2$) of these two tensile stresses with allowable stress σ for the stud material.

$$\sigma_1+\sigma_2<\sigma$$

$$\sigma_1=\frac{F_r\cdot\ell}{Z}$$

F_r : Maximum radial load
Z : Coefficient of shaft section through Point-A

$\sigma_2 \fallingdotseq$ 98MPa (10kgf/mm²)
Tensile stress generating from maximum tightening torque described in "Dimensions Table"

σ : Allowable stress for material
The following values are adopted from the repeated bending test result of the stud material.
Where the stud material is subjected to static bending stress;
σ = 1372MPa (140kgf/mm²)
Where the stud material is subjected to repeated bending stress (single direction)
σ = 784MPa (80kgf/mm²)
Where the stud material is subjected to repeated bending stress (double directions)
σ = 392MPa (40kgf/mm²)

Accordingly,

$$F_r<\frac{Z}{\ell}(\sigma-\sigma_2)$$

12. Bearing Type Symbols and Auxiliary Symbols

Table 12.1 Bearing Type Symbols

Type code	Bearing type
811	Single-direction thrust cylindrical roller bearing, dimension series 11
812	Single-direction thrust cylindrical roller bearing, dimension series 12
893	Single-direction thrust cylindrical roller bearing, dimension series 93
A	Needle roller, spherical type
AS11	Steel plate thrust washer, dimension series 11
ARN	Needle roller bearing with double-direction thrust cylindrical roller bearing
AXK11	Thrust needle roller bearing, dimension series 11
AXN	Needle roller bearing with double-direction thrust needle roller bearing
BF	Metallic flat cage for linear flat rollers
BK	Drawn cup needle roller bearing with close end
BR	Snap ring for bearing housing
CRV	Full complement roller for cam follower,
DCL	inch series
	Drawn cup needle roller bearing with open end, inch series
F	Needle roller, plane type
FF	Linear flat roller
FR	Bottom roller, for drawing frame (textile machine)
FRIS	Bottom roller
G	Synthetic rubber seal, one-lip type
GD	Synthetic rubber seal, double-lip type
GK	Needle rollers with split type cage
GS811	Outer ring for thrust bearing, dimension series 11
GS812	Outer ring for thrust bearing, dimension series 12
GS893	Outer ring for thrust bearing, dimension series 93
HCK	Drawn cup needle roller bearing for cross joint
HF	One-way clutch
HFL	One-way clutch integral with bearing
HK	Drawn cup needle roller bearing with open end
HMK	Drawn cup needle roller bearing with open end, for heavy load application
IR	Inner ring
JF··S	Arms for tension pulley and jockey pulley
JPU··S	Tension pulley and jockey pulley
K	Needle rollers with cage
K811	Thrust cylindrical roller, dimension series 11
K812	Thrust cylindrical roller, dimension series 12
K893	Thrust cylindrical roller, dimension series 93
KBK	Needle roller and cage assembly for small ends
KD	Linear ball bearing, stroking type
KH	Linear ball bearing, drawn cup type
KLM	Linear ball bearing, machined ring type
KLM··S	Linear ball bearing, clearance-adjustable type
KLM··P	Linear ball bearing, open type
KR	Cam follower
KRU	Cam follower, shaft eccentric type
KRV	Cam follower, full complement roller type
KRVU	Cam follower, full complement roller and shaft eccentric type

Type code	Bearing type
MI	Inner ring, inch series
MR	Machined ring needle roller bearing without inner ring, inch series
NA22	Roller follower with inner ring, dimension series 22
NA48	Machined ring needle roller bearing with inner ring, dimension series 48
NA49	Machined ring needle roller bearing with inner ring, dimension series 49
NA59	Machined ring needle roller bearing with inner ring, dimension series 59
NA69	Machined ring needle roller bearing with inner ring, dimension series 69
NA49···S	Clearance-adjustable needle roller bearing with inner ring
NAO	Machined ring needle roller bearing, separable type, with inner ring
NATR	Roller follower
NATV	Roller follower, full complement roller type
NIP	Grease nipple
NK	Machined ring needle roller bearing without inner ring
NKIA59	Complex bearing : Needle roller bearing with angular ball bearing dimension series 59 Complex bearing
NKIB59	Needle roller bearing with three-point contact type ball bearing dimension series 59
NKX	Complex bearing : needle roller bearing with thrust ball bearing without dust-proof cover
NKX···Z	Complex bearing: Needle roller bearing with thrust ball bearing with dust-proof cover
NKXR	Complex bearing: Needle roller bearing with thrust cylindrical roller bearing without dust-proof cover
NKXR···Z	Complex bearing: Needle roller bearing with thrust cylindrical roller bearing with dust-proof cover
NUKR	Cam follower, full complement roller type
NUTR2	Roller follower, diameter series 2
NUTR33	Roller follower, diameter series 3
NUTW	Roller follower outer ring, middle rib type
PK	Needle roller and cage assembly
RF	Polyamide resin cage for linear flat rollers
RLM	Linear roller bearing
RNA22	Roller follower without inner ring, dimension series 22
RNA48	Machined ring needle roller bearing without inner ring, dimension series 48
RNA49	Machined ring needle roller bearing without inner ring, dimension series 49
RNA59	Machined ring needle roller bearing without inner ring, dimension series 59
RNA69	Machined ring needle roller bearing without inner ring, dimension series 69
RNA49···S	Clearance-adjustable needle roller bearing, without inner ring
RNAB2	Open type roller follower without inner ring, diameter series 2
RNAO	Machined ring needle roller bearing, separable type, without inner ring
WR	Snap ring for shaft
WS811	Thrust inner ring, dimension series 11
WS812	Thrust inner ring, dimension series 12
WS893	Thrust inner ring, dimension series 93

Table 1.2.2 Auxiliary symbols

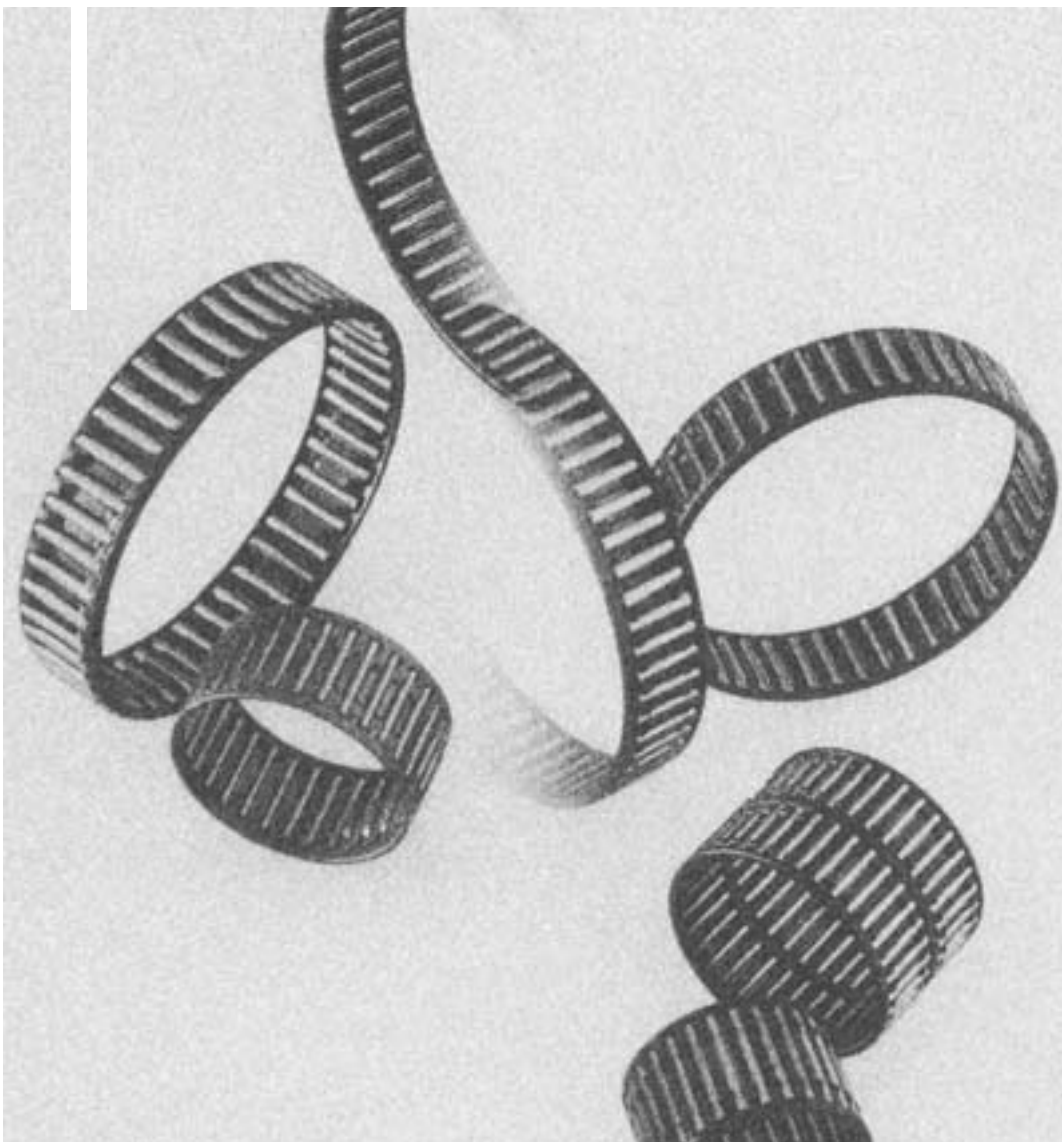
	Symbol		Symbol representation
Initial symbols	Material heat-treatment symbols	TS-	Bearing for high temperature application which was heat-treated for dimensional stabilization
		M-	Plated bearing
		E-	Bearing made of case-hardened steel
		F-	Bearing made of stainless steel
		H-	Bearing made of high speed steel
		C-	Bearing made of carbon steel
	Expansion compensation	EC-	Expansion-compensated bearing
Basic symbols			
Tail codes	Internal construction symbols	ZW	Double-row cage
		A,B,C	Internal construction change
	Cage symbols	J	Steel plate punched cage
		F1	Machined iron cage
		L1	High-strengh yellow copper cage
		T2	Polyamide resin cage
		L3	Aluminum alloy cage
		L5	Sintered alloy cage
		S	Welded cage
	Seal symbols	L,LL	With synthetic rubber seal
		P,PP	With plastic seal
	Bearing ring profile symbols	K	Bearing with 1/12 tapered bore
		N	Grooved snap ring
		NR	With snap ring
		W	Bearing ring with notched knock hole
		D	With oil hole
		D1	With oil hole and oil groove
		H	Cam follower with hexagon hole
	Combination symbols	D2,Dn	Complex bearing using two or more same bearings
		+α	With spacer
		-P	Without cross roller stud
	Clearance symbols	C2	Clearance smaller than ordinary clearance
		C3	Clearance larger than ordinary clearance
		C4	Radial clearance larger than C3
		NA	Non-interchangeable clearance
	Accuracy class symbols	P6	Bearing of JIS Class-6
		P5	Bearing of JIS Class-5
		P4	Bearing of JIS Class-4
	Lubrication symbols	/2A	SHELL ALVANIA Grease 2
		/3A	SHELL ALVANIA Grease 3
	Special symbols	V1～Vn	Special specification, requirements

Needle Roller Bearings

BEARING TABLES

Needle roller and cage assemblies	B- 3～ 20
Needle roller and cage assemblies for connecting rod bearings	B- 21～ 28
Drawn cup needle roller bearings	B- 29～ 52
Machined-ring needle roller bearings	B- 53～ 96
Machined-ring needle roller bearings, separable	B- 97～114
Inner rings	B-115～132
Clearance-adjustable needle roller bearings	B-133～138
Complex bearings	B-139～160
Roller followers	B-161～173
Cam followers	B-174～204
Thrust roller bearings	B-205～220
Components Needle rollers / Snap rings / Seals	B-221～234
Linear bearings	B-235～254
One-way clutches	B-255～259
Bottom roller bearings for textile machinery Tension pulleys for textile machinery	B-260～266

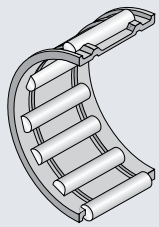
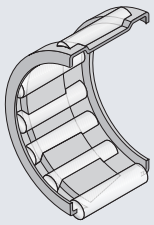
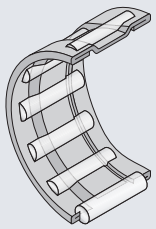
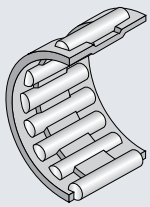
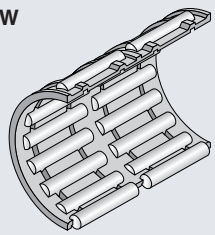
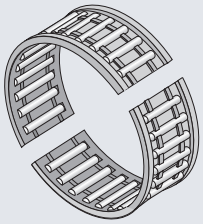
Needle Roller and Cage Assemblies



Needle Roller and Cage Assemblies

This needle roller and cage assembly is one of the basic components for the needle roller bearing of a construction wherein the needle rollers are fitted with a cage so as not to separate from each other. The use of this roller and cage assembly enables to design a compact and lightweight construction by utilizing a shaft or a housing as the direct raceway surface, without using inner ring and outer ring.

The needle rollers are guided by the cage more precisely than the full complement roller type, hence enabling high speed running of bearing. In addition, its rigidity and load capacity are high for the specific space.

Type of needle roller and cage assembly	Cage type	Applied shaft diameter (mm)	Composition of nominal bearing number	Nominal bearing number	Code items and respective dimensions
K **(K··S)** **(K··T2)** 	Machined ring type Polyamide resin type	ϕ3～ϕ285	**K 20 × 24 × 10 S** S: Tail code 10: Width 24: Roller set outer diameter 20: Inscribed circle diameter K: Type code	K20×24×10S	Inscribed circle diameter : ϕ20 Roller set outer diameter : ϕ24 Width : 10 S : Welded type
	Welded type	ϕ10～ϕ100			
KMJ **(KMJ··S)** **(PCJ)** 	Punched type	ϕ15～ϕ100	**KMJ 20 × 26 × 13** 13: Width 26: Roller set outer diameter 20: Inscribed circle diameter KMJ: Type code	KMJ20×26×13	Inscribed circle diameter : ϕ20 Roller set outer diameter : ϕ26 Width : 13
	Welded type	ϕ10～ϕ40			
KJ··S 	Welded type	ϕ20～ϕ40	**KJ 30 × 35 × 17 S** S: Tail code 17: Width 35: Roller set outer diameter 30: Inscribed circle diameter KJ: Type code	KJ30×35×17S	Inscribed circle diameter : ϕ30 Roller set outer diameter : ϕ35 Width : 17 S : Welded type
KV··S 	Welded type	ϕ7～ϕ100	**KV 30 × 35 × 17 S** S: Tail code 17: Width 35: Roller set outer diameter 30: Inscribed circle diameter KV: Type code	KV30×35×17S	Inscribed circle diameter : ϕ30 Roller set outer diameter : ϕ35 Width : 17 S : Welded type
K··ZW 	Machined ring type	ϕ8～ϕ285	**K 20 × 24 × 45 ZW** ZW: Tail code 45: Width 24: Roller set outer diameter 20: Inscribed circle diameter K: Type code	K20×24×45ZW	Inscribed circle diameter : ϕ20 Roller set outer diameter : ϕ24 Width : 45 ZW : Double-row type
GK 	Machined ring type Split type	ϕ8～ϕ285	**GK 30 × 35 × 17** 17: Width 35: Roller set outer diameter 30: Inscribed circle diameter GK: Type code	GK30×35×17	Inscribed circle diameter : ϕ30 Roller set outer diameter : ϕ35 Width : 17

The ones with tail code T2 using polyamide resin cage shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C or less.
The applied shaft diameter of welded type are described based on size that manufacturers can deal with a manufacturing. Therefore, it has nothing to do with the Demention Table.

Inter-roller diameter deviation of needle rollers

Inter-roller diameter deviation of needle rollers which are built in one cage is within 2um. The respective specified diameter tolerances for built-in needle rollers are identified by identification colors on the package label, as shown in **Table 1**.

The standard identification colors marked on each label are red, dark blue, blue, black and white, unless otherwise specially indicated.

Where two or more bearings are used in tandem, those of same identification color must be used for equal distribution of bearing load.

Table 1 Color identification of diameter tolerances for needle roller bearings

Identification color on label	Diameter tolerance μm	Classification
Red	0～－ 2	Standard
Dark blue	－1～－ 3	
Blue	－2～－ 4	
Black	－3～－ 5	
White	－4～－ 6	
Gray	－5～－ 7	Semi-standard
Green	－6～－ 8	
Brown	－7～－ 9	
Yellow	－8～－10	

Radial Clearance

When a shaft and a housing are used as the direct raceway surface, the radial clearance is determined by the shaft diameter and the housing hole dimension. Where the diameter tolerances for needle rollers are standard (tolerances color-identified in red, dark blue, blue, black, white), the shaft diameters and housing hole dimensions shown in **Table 2** are used.

Table 2 Recommended fits

Shaft diameter mm		～80	80～140	140～
Radial clearance	Smaller than ordinary clearance	j5/G6	h5/G6	h5/G6
	Ordinary clearance	h5/G6	g5/G6	f5/H6
	Larger than ordinary clearance	g6/G6	f6/G6	f6/G6

Shaft and housing specifications

Where a shaft and a housing are used as the direct raceway surface, the raceway surface must meet the specifications described in **Table 3**.

Table 3 Shaft and housing specifications

Characteristics	Shaft	Housing
Roundness (max)	IT3	IT4
Cylindricality (max)	IT3	IT4
Surface roughness (max)	Refer to **Table 7.4** on page A-39.	
Surface hardness	HRC58～64	
Depth of hardened layer (min)	Refer to formula **(7.1)** on page A-39	

Bearing installation related dimensions

The tolerances for the nominal dimension "Bc" of cage specified in **Table 4** are recommended as the guide width b of cage.

Table 4 Guide width tolerances

Unit : mm

Classification	Tolerances for guide width b	
	Single row	Double row
Metric system	$Bc\ {}^{+0.2}_{0}$	$Bc\ {}^{+0.3}_{+0.1}$
Inch system	$Bc\ {}^{+0.4}_{+0.2}$	$Bc\ {}^{+0.5}_{+0.3}$

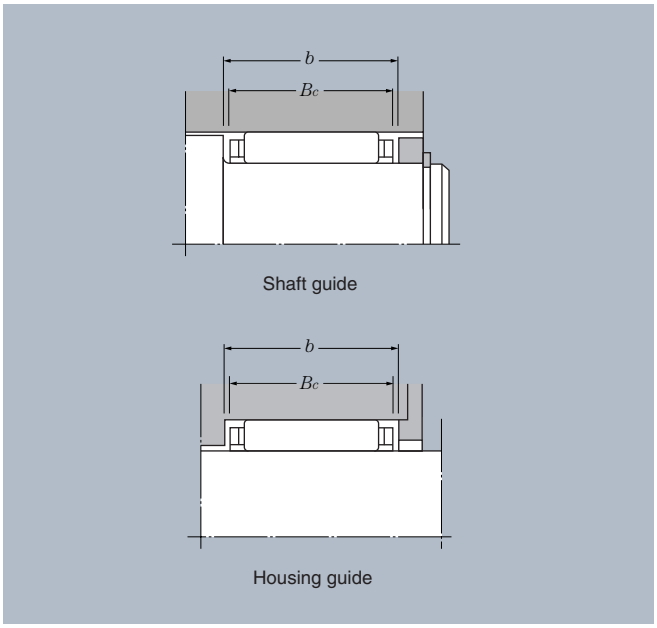
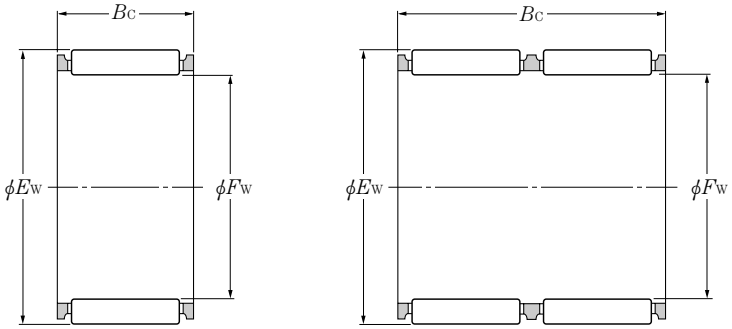


Fig. 1

Type K
Type K··T2
Type K··S
Type K··ZW
Type KMJ



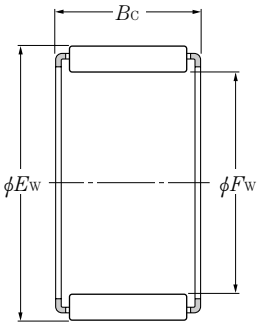
Type K
Type K··T2
Type K··S

Type K··ZW

F_w 3～12mm

Boundary dimensions mm			Basic load ratings N dynamic		kgf dynamic		Limiting speeds r/min grease	oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
3	6	7 $^{-0.2}_{-0.55}$	1 460	970	149	99	33 000	50 000	**K3×6×7T2**	0.0004
4	7	7 $^{-0.2}_{-0.55}$	1 770	1 270	180	129	30 000	45 000	**K4×7×7T2**	0.0005
5	8	8 $^{-0.2}_{-0.55}$	2 640	2 190	269	224	27 000	40 000	**K5×8×8T2**	0.0007
	8	10	2 720	2 250	277	230	27 000	40 000	**K5×8×10T2**	0.0009
6	9	8 $^{-0.2}_{-0.55}$	2 660	2 280	272	233	25 000	37 000	**K6×9×8T2**	0.0009
	9	10	3 400	3 150	345	320	25 000	37 000	**K6×9×10T2**	0.0011
	10	13	4 400	3 700	450	380	25 000	37 000	**K6×10×13T2**	0.0019
7	10	8 $^{-0.2}_{-0.55}$	2 670	2 350	272	239	23 000	34 000	**K7×10×8T2**	0.0009
	10	10	3 400	3 200	345	330	23 000	34 000	**K7×10×10T2**	0.0011
8	11	8 $^{-0.2}_{-0.55}$	3 150	3 000	320	305	21 000	32 000	**K8×11×8T2**	0.0011
	11	10	4 000	4 100	410	420	21 000	32 000	**K8×11×10T2**	0.0013
	11	13	4 850	5 200	495	535	21 000	32 000	**K8×11×13**	0.0026
	12	10	4 650	4 150	475	425	21 000	32 000	**K8×12×10T2**	0.0020
	12	12	5 600	5 300	570	540	21 000	32 000	**K8×12×12**	0.0034
	12	13	5 600	5 300	570	540	21 000	32 000	**K8×12×13**	0.0036
9	12	10 $^{-0.2}_{-0.55}$	4 550	5 000	465	510	20 000	30 000	**K9×12×10T2**	0.0015
	12	13	5 500	6 400	560	650	20 000	30 000	**K9×12×13T2**	0.0021
10	13	10 $^{-0.2}_{-0.55}$	4 550	5 100	460	520	19 000	28 000	**K10×13×10T2**	0.0016
	13	13	5 450	6 450	555	660	19 000	28 000	**K10×13×13**	0.0031
	14	8	4 300	3 950	435	405	19 000	28 000	**K10×14×8**	0.0027
	14	10	5 500	5 450	560	555	19 000	28 000	**K10×14×10**	0.0034
	14	13	6 600	6 900	675	705	19 000	28 000	**K10×14×13**	0.0044
	16	12	7 100	5 950	720	610	19 000	28 000	**K10×16×12**	0.0066
11	14	10 $^{-0.2}_{-0.55}$	5 050	6 000	515	615	18 000	27 000	**K11×14×10**	0.0028
12	15	9 $^{-0.2}_{-0.55}$	4 450	5 250	455	535	17 000	26 000	**K12×15×9**	0.0027
	15	10	5 000	6 100	510	620	17 000	26 000	**K12×15×10S**	0.0033
	15	13	6 000	7 700	615	785	17 000	26 000	**K12×15×13**	0.0038
	15	20	8 550	12 200	875	1 240	17 000	26 000	**K12×15×20ZW**	0.0059
	16	8	4 850	4 900	495	500	17 000	26 000	**K12×16×8**	0.0034

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.



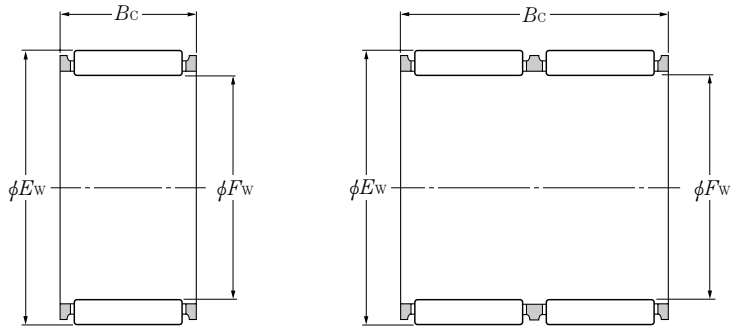
Type KMJ

F_w 12～16mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
	16	13	7 500	8 500	765	870	17 000	26 000	**K12×16×13**	0.0055
12	17	13 $^{-0.2}_{-0.55}$	9 000	9 400	920	960	17 000	26 000	**K12×17×13**	0.0075
	18	12	8 650	8 000	880	815	17 000	26 000	**K12×18×12**	0.0084
	17	10	5 400	7 050	550	720	16 000	24 000	**K14×17×10**	0.0033
	18	10	6 900	8 000	705	815	16 000	24 000	**K14×18×10**	0.0046
	18	11	7 600	9 050	775	925	16 000	24 000	**K14×18×11**	0.0053
	18	13	8 300	10 100	845	1 030	16 000	24 000	**K14×18×13**	0.0063
	18	15 $^{-0.2}_{-0.55}$	9 650	12 300	985	1 250	16 000	24 000	**K14×18×15S**	0.0076
14	18	17	10 900	14 400	1 120	1 470	16 000	24 000	**K14×18×17**	0.0081
	18	39	18 800	28 900	1 910	2 950	16 000	24 000	**K14×18×39ZW**	0.0180
	19	13	8 950	9 650	915	985	16 000	24 000	**K14×19×13**	0.0080
	20	12	9 350	9 150	955	930	16 000	24 000	**K14×20×12**	0.0095
	20	17	13 500	14 600	1 370	1 490	16 000	24 000	**K14×20×17**	0.0140
	18	14	7 850	11 600	800	1 190	15 000	23 000	**K15×18×14**	0.0060
	19	10	6 850	8 050	700	820	15 000	23 000	**K15×19×10S**	0.0051
	19	13	8 250	10 200	840	1 040	15 000	23 000	**K15×19×13S**	0.0067
	19	17	10 900	14 600	1 110	1 490	15 000	23 000	**K15×19×17**	0.0090
15	19	24 $^{-0.2}_{-0.55}$	14 100	20 400	1 440	2 080	15 000	23 000	**K15×19×24ZW**	0.0130
	20	13	10 100	11 500	1 030	1 170	15 000	23 000	**K15×20×13**	0.0088
	21	15	11 900	12 500	1 210	1 280	15 000	23 000	**K15×21×15**	0.0130
	21	17	15 100	17 200	1 540	1 750	15 000	23 000	**KMJ15×21×17**	0.0140
	21	21	16 500	19 100	1 680	1 950	15 000	23 000	**K15×21×21**	0.0170
	20	10	7 500	9 250	765	945	15 000	23 000	**K16×20×10**	0.0057
	20	11	8 300	10 500	845	1 070	15 000	23 000	**K16×20×11**	0.0061
	20	13	9 050	11 800	925	1 200	15 000	23 000	**K16×20×13**	0.0071
	20	17 $^{-0.2}_{-0.55}$	11 900	16 800	1 220	1 710	15 000	23 000	**K16×20×17S**	0.0092
16	22	12	11 700	12 500	1 190	1 280	15 000	23 000	**K16×22×12**	0.0100
	22	13	12 600	13 900	1 290	1 410	15 000	23 000	**KMJ16×22×13**	0.0110
	22	16	14 200	16 100	1 450	1 650	15 000	23 000	**KMJ16×22×16**	0.0140
	22	17	14 400	16 400	1 470	1 670	15 000	23 000	**K16×22×17**	0.0150

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

Type K
Type K··S◦
Type K··ZW
Type KMJ
Type KMJ··S◦



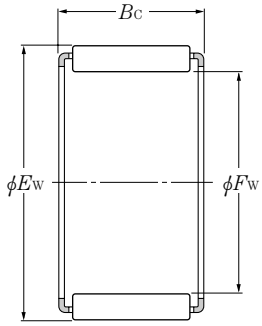
Type K
Type K··S

Type K··ZW

F_w 16～20mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	oil	Bearing numbers	Mass kg (approx.)
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				
16	22	20 $^{-0.2}_{-0.55}$	16 000	18 800	1 640	1 920	15 000	23 000	**K16×22×20**	0.0170
17	21	10	7 450	9 300	760	950	15 000	22 000	**K17×21×10S**	0.0056
	21	13	9 400	12 600	960	1 280	15 000	22 000	**K17×21×13S**	0.0075
	21	15 $^{-0.2}_{-0.55}$	10 400	14 400	1 060	1 460	15 000	22 000	**K17×21×15**	0.0089
	21	17 $^{-0.2}_{-0.55}$	11 800	16 900	1 210	1 720	15 000	22 000	**K17×21×17**	0.0095
	22	20	14 700	19 200	1 500	1 960	15 000	22 000	**K17×22×20**	0.0150
	23	17	14 400	16 500	1 460	1 690	15 000	22 000	**K17×23×17**	0.0160
18	22	10	7 400	9 400	755	955	14 000	21 000	**K18×22×10**	0.0061
	22	13	8 900	11 900	910	1 210	14 000	21 000	**K18×22×13**	0.0077
	22	17	11 700	17 000	1 200	1 730	14 000	21 000	**K18×22×17**	0.0110
	23	20	14 600	19 300	1 490	1 970	14 000	21 000	**K18×23×20S**	0.0150
	24	12 $^{-0.2}_{-0.55}$	12 300	13 800	1 250	1 410	14 000	21 000	**K18×24×12**	0.0120
	24	13 $^{-0.2}_{-0.55}$	11 600	12 800	1 180	1 300	14 000	21 000	**K18×24×13**	0.0130
	24	17	16 000	19 300	1 630	1 970	14 000	21 000	**KMJ18×24×17S**	0.0140
	24	20	17 000	20 900	1 730	2 130	14 000	21 000	**K18×24×20**	0.0190
	25	17	18 000	20 400	1 830	2 080	14 000	21 000	**K18×25×17**	0.0190
	25	22	22 100	26 600	2 250	2 710	14 000	21 000	**K18×25×22**	0.0240
19	23	13 $^{-0.2}_{-0.55}$	9 650	13 500	985	1 370	14 000	21 000	**K19×23×13**	0.0082
	23	17 $^{-0.2}_{-0.55}$	12 700	19 200	1 300	1 960	14 000	21 000	**K19×23×17**	0.0110
20	24	10	8 300	11 200	845	1 150	13 000	20 000	**K20×24×10S**	0.0065
	24	11	9 500	13 400	970	1 370	13 000	20 000	**K20×24×11**	0.0072
	24	13	10 000	14 300	1 020	1 460	13 000	20 000	**K20×24×13S**	0.0086
	24	17	13 200	20 400	1 340	2 080	13 000	20 000	**K20×24×17S**	0.0110
	24	45	16 400	27 100	1 680	2 760	13 000	20 000	**K20×24×45ZW**	0.0280
	25	40 $^{-0.2}_{-0.55}$	29 000	48 000	2 950	4 900	13 000	20 000	**K20×25×40ZW**	0.0330
	26	12	12 900	15 100	1 320	1 540	13 000	20 000	**K20×26×12**	0.0130
	26	13	14 000	16 700	1 420	1 700	13 000	20 000	**KMJ20×26×13**	0.0120
	26	17	17 800	22 800	1 810	2 330	13 000	20 000	**KMJ20×26×17S**	0.0160
	26	20	20 600	27 600	2 100	2 820	13 000	20 000	**KMJ20×26×20S**	0.0190
	28	25	27 100	32 500	2 760	3 300	13 000	20 000	**K20×28×25**	0.0370

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.



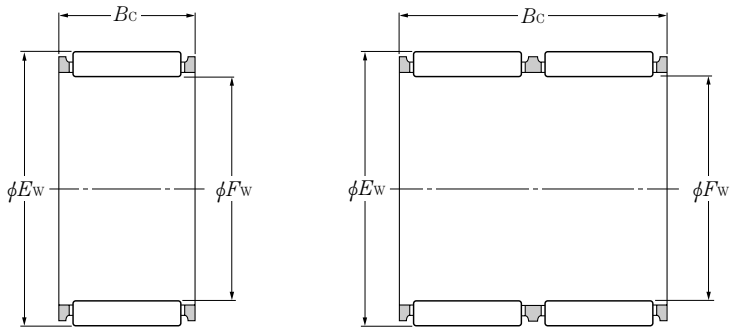
Type KMJ
Type KMJ··S

F_W 21～25mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
21	25	13 $^{-0.2}_{-0.55}$	10 300	15 100	1 050	1 540	13 000	19 000	**K21×25×13**	0.0090
	25	17 $^{-0.2}_{-0.55}$	13 600	21 500	1 380	2 200	13 000	19 000	**K21×25×17**	0.0120
22	26	10	8 500	11 900	865	1 220	12 000	18 000	**K22×26×10S**	0.0071
	26	11	10 100	14 900	1 030	1 520	12 000	18 000	**K22×26×11S**	0.0079
	26	13	10 200	15 200	1 040	1 550	12 000	18 000	**K22×26×13**	0.0094
	26	17	13 500	21 600	1 370	2 200	12 000	18 000	**K22×26×17S**	0.0120
	27	20 $^{-0.2}_{-0.55}$	17 500	25 900	1 780	2 640	12 000	18 000	**K22×27×20**	0.0200
	27	40	29 900	51 500	3 050	5 250	12 000	18 000	**K22×27×40ZW**	0.0390
	28	17	17 700	23 300	1 810	2 380	12 000	18 000	**K22×28×17**	0.0200
	29	16	18 700	22 700	1 910	2 310	12 000	18 000	**KMJ22×29×16**	0.0220
	30	15	19 300	21 700	1 970	2 210	12 000	18 000	**K22×30×15**	0.0220
23	28	24 $^{-0.2}_{-0.55}$	19 800	31 000	2 020	3 150	11 000	17 000	**K23×28×24**	0.0230
24	28	10	9 000	13 200	915	1 350	11 000	17 000	**K24×28×10S**	0.0081
	28	13	10 800	16 800	1 100	1 710	11 000	17 000	**K24×28×13**	0.0100
	28	17 $^{-0.2}_{-0.55}$	14 300	23 900	1 460	2 440	11 000	17 000	**K24×28×17**	0.0130
	29	13	12 300	16 900	1 250	1 720	11 000	17 000	**K24×29×13**	0.0120
	30	17	18 400	25 200	1 880	2 570	11 000	17 000	**K24×30×17**	0.0220
	30	31	27 900	43 000	2 840	4 350	11 000	17 000	**K24×30×31ZW**	0.0390
25	29	10	8 950	13 300	910	1 350	11 000	16 000	**K25×29×10**	0.0083
	29	13	10 800	16 900	1 100	1 720	11 000	16 000	**K25×29×13**	0.0100
	29	17	14 200	24 000	1 450	2 450	11 000	16 000	**K25×29×17S**	0.0140
	30	13	13 200	18 800	1 350	1 920	11 000	16 000	**K25×30×13**	0.0130
	30	17	17 400	26 800	1 770	2 730	11 000	16 000	**K25×30×17S**	0.0170
	30	20 $^{-0.2}_{-0.55}$	19 400	31 000	1 980	3 150	11 000	16 000	**K25×30×20S**	0.0210
	30	26	21 800	35 500	2 220	3 650	11 000	16 000	**K25×30×26ZW**	0.0270
	30	39	29 800	53 500	3 050	5 450	11 000	16 000	**K25×30×39ZW**	0.0400
	31	13	15 200	19 900	1 550	2 030	11 000	16 000	**K25×31×13**	0.0160
	31	14	16 500	22 100	1 680	2 250	11 000	16 000	**K25×31×14**	0.0180
	31	17	18 300	25 300	1 870	2 580	11 000	16 000	**K25×31×17**	0.0220
	31	21	22 500	33 000	2 290	3 350	11 000	16 000	**K25×31×21**	0.0260

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

Type K
Type K··S◦
Type K··ZW
Type KMJ
Type KMJ··S◦
Type KJ··S◦



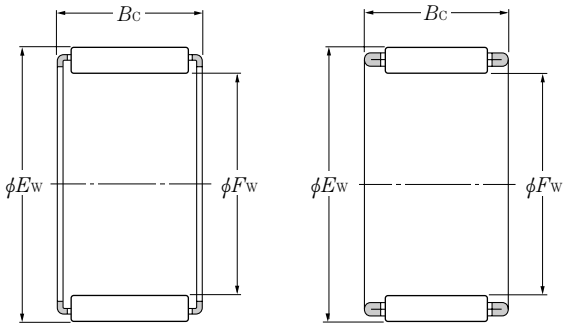
Type K
Type K··S

Type K··ZW

F_w 25～32mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
25	32	16 $^{-0.2}_{-0.55}$	19 500	24 700	1 990	2 520	11 000	16 000	**KMJ25×32×16**	0.025
	33	24 $^{-0.2}_{-0.55}$	34 500	47 000	3 500	4 800	11 000	16 000	**KMJ25×33×24S**	0.040
26	30	13	11 800	19 200	1 200	1 960	10 000	15 000	**K26×30×13**	0.011
	30	17 $^{-0.2}_{-0.55}$	15 500	27 400	1 580	2 790	10 000	15 000	**K26×30×17**	0.015
	34	22	24 200	30 000	2 470	3 050	10 000	15 000	**K26×34×22**	0.041
28	32	17	15 300	27 500	1 560	2 810	9 500	14 000	**K28×32×17**	0.017
	32	21	18 700	35 500	1 910	3 650	9 500	14 000	**K28×32×21**	0.020
	33	13	13 900	20 900	1 420	2 130	9 500	14 000	**K28×33×13**	0.015
	33	17	18 300	29 800	1 870	3 050	9 500	14 000	**K28×33×17S**	0.020
	33	26 $^{-0.2}_{-0.55}$	23 900	42 000	2 430	4 250	9 500	14 000	**K28×33×26ZW**	0.033
	33	27 $^{-0.2}_{-0.55}$	28 300	52 000	2 890	5 300	9 500	14 000	**K28×33×27**	0.032
	34	14	17 500	24 800	1 790	2 530	9 500	14 000	**K28×34×14**	0.020
	34	17	18 100	25 800	1 850	2 630	9 500	14 000	**K28×34×17**	0.024
	35	16	21 200	28 400	2 160	2 900	9 500	14 000	**K28×35×16**	0.029
	35	18	21 500	28 900	2 190	2 950	9 500	14 000	**K28×35×18**	0.031
29	34	17 $^{-0.2}_{-0.55}$	18 900	31 000	1 920	3 200	9 500	14 000	**K29×34×17S**	0.022
	34	27 $^{-0.2}_{-0.55}$	28 100	52 000	2 870	5 300	9 500	14 000	**K29×34×27**	0.033
30	35	11	12 200	18 000	1 240	1 840	8 500	13 000	**K30×35×11S**	0.014
	35	13	14 700	22 900	1 500	2 340	8 500	13 000	**K30×35×13S**	0.016
	35	17	19 400	32 500	1 970	3 350	8 500	13 000	**KJ30×35×17S**	0.021
	35	20	21 600	37 500	2 200	3 850	8 500	13 000	**K30×35×20S**	0.025
	35	26	25 200	46 000	2 570	4 650	8 500	13 000	**K30×35×26ZW**	0.032
	35	27 $^{-0.2}_{-0.55}$	29 900	57 000	3 050	5 800	8 500	13 000	**K30×35×27S**	0.033
	36	14 $^{-0.2}_{-0.55}$	18 600	27 400	1 900	2 790	8 500	13 000	**KMJ30×36×14**	0.022
	37	16	21 900	30 500	2 230	3 100	8 500	13 000	**K30×37×16**	0.029
	37	18	23 300	33 000	2 370	3 350	8 500	13 000	**K30×37×18**	0.034
	37	20	26 200	38 000	2 670	3 850	8 500	13 000	**KMJ30×37×20**	0.038
	37	48	40 000	65 500	4 050	6 700	8 500	13 000	**K30×37×48ZW**	0.075
	38	18	25 000	33 000	2 550	3 350	8 500	13 000	**K30×38×18**	0.036
32	36	15 $^{-0.2}_{-0.55}$	14 300	26 400	1 460	2 700	8 500	13 000	**K32×36×15S**	0.017

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.



Type KMJ
Type KMJ··S

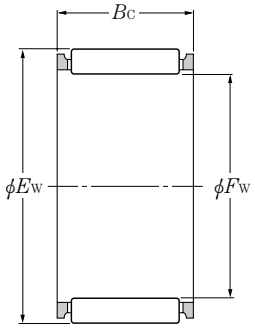
Type KJ··S

F_W 32～38mm

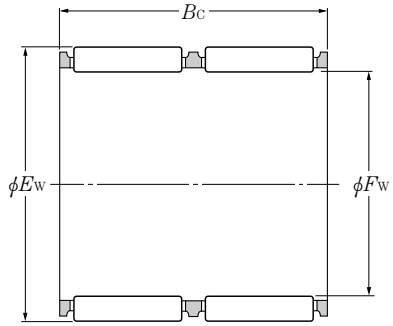
Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
32	37	13	14 500	23 000	1 480	2 350	8 500	13 000	**K32×37×13**	0.018
	37	17	19 200	33 000	1 950	3 350	8 500	13 000	**K32×37×17S**	0.022
	37	26	24 900	46 000	2 540	4 700	8 500	13 000	**K32×37×26ZW**	0.034
	37	27	29 600	57 500	3 000	5 850	8 500	13 000	**KJ32×37×27S**	0.037
	38	14 $^{-0.2}_{-0.55}$	19 800	30 500	2 020	3 100	8 500	13 000	**KMJ32×38×14**	0.022
	38	20	25 100	41 000	2 560	4 150	8 500	13 000	**KJ32×38×20S**	0.031
	38	26	31 500	54 000	3 200	5 550	8 500	13 000	**K32×38×26**	0.041
	39	16	22 600	32 000	2 310	3 300	8 500	13 000	**K32×39×16**	0.034
	39	18	24 000	35 000	2 450	3 550	8 500	13 000	**K32×39×18**	0.037
	39	20	26 800	40 000	2 740	4 100	8 500	13 000	**KJ32×39×20S**	0.041
35	40	13	15 200	25 100	1 550	2 560	7 500	11 000	**K35×40×13**	0.019
	40	17	20 000	36 000	2 040	3 650	7 500	11 000	**K35×40×17**	0.025
	40	26	26 100	50 000	2 660	5 100	7 500	11 000	**K35×40×26ZW**	0.037
	40	27	32 000	65 000	3 250	6 600	7 500	11 000	**KJ35×40×27S**	0.039
	40	30	26 100	50 000	2 660	5 100	7 500	11 000	**K35×40×30ZW**	0.043
	41	14	19 400	30 500	1 980	3 100	7 500	11 000	**K35×41×14**	0.026
	41	15 $^{-0.2}_{-0.55}$	20 900	33 500	2 130	3 400	7 500	11 000	**K35×41×15**	0.027
	41	40	43 000	84 000	4 400	8 550	7 500	11 000	**K35×41×40ZW**	0.055
	42	16	24 100	36 000	2 450	3 650	7 500	11 000	**K35×42×16**	0.035
	42	18	24 700	37 000	2 510	3 750	7 500	11 000	**K35×42×18**	0.039
	42	20	28 500	44 500	2 910	4 550	7 500	11 000	**K35×42×20**	0.041
	42	30	39 500	68 000	4 050	6 950	7 500	11 000	**K35×42×30**	0.062
	42	45	42 500	74 000	4 300	7 550	7 500	11 000	**K35×42×45ZW**	0.106
36	42	46 $^{-0.2}_{-0.55}$	51 000	106 000	5 200	10 800	7 500	11 000	**K36×42×46ZW**	0.086
37	42	13	15 900	27 100	1 620	2 770	7 500	11 000	**K37×42×13**	0.021
	42	17 $^{-0.2}_{-0.55}$	20 400	37 500	2 080	3 800	7 500	11 000	**KJ37×42×17S**	0.026
	42	27	32 500	67 500	3 300	6 900	7 500	11 000	**KJ37×42×27S**	0.041
	44	18	26 300	41 000	2 680	4 150	7 500	11 000	**K37×44×18**	0.042
38	43	17 $^{-0.2}_{-0.55}$	20 900	38 500	2 130	3 950	7 500	11 000	**K38×43×17**	0.026
	43	27	32 000	67 500	3 300	6 900	7 500	11 000	**K38×43×27**	0.043

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.

Type K
Type K··ZW
Type KMJ
Type KJ··S◦



Type K

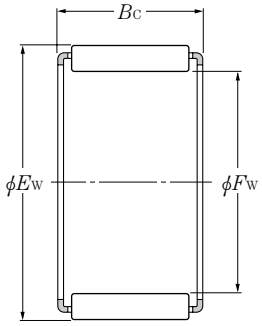


Type K··ZW

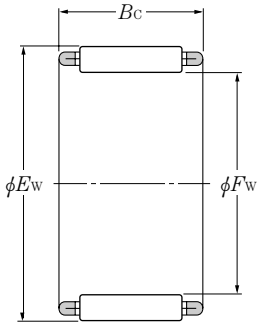
F_W 38～48mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_W	E_W	B_C	C_r	C_{or}	C_r	C_{or}				(approx.)
38	46	20 $^{-0.2}_{-0.55}$	34 000	52 000	3 450	5 350	7 500	11 000	**KMJ38×46×20**	0.046
	46	32 $^{-0.2}_{-0.55}$	54 000	95 500	5 500	9 700	7 500	11 000	**K38×46×32**	0.073
40	45	13	16 500	29 200	1 680	2 980	6 500	10 000	**K40×45×13**	0.022
	45	17	21 800	41 500	2 220	4 250	6 500	10 000	**K40×45×17**	0.027
	45	21	26 700	54 000	2 720	5 500	6 500	10 000	**K40×45×21**	0.037
	45	27	33 500	72 500	3 400	7 400	6 500	10 000	**K40×45×27**	0.044
	46	17 $^{-0.2}_{-0.55}$	24 600	43 000	2 500	4 350	6 500	10 000	**K40×46×17**	0.030
	47	18	27 700	45 000	2 820	4 550	6 500	10 000	**K40×47×18**	0.045
	47	20	31 000	51 500	3 150	5 250	6 500	10 000	**K40×47×20**	0.048
	48	20	33 000	51 000	3 350	5 200	6 500	10 000	**K40×48×20**	0.052
	48	25	42 500	71 000	4 350	7 250	6 500	10 000	**K40×48×25**	0.065
42	47	17	22 100	43 000	2 250	4 400	6 500	9 500	**K42×47×17**	0.028
	47	27 $^{-0.2}_{-0.55}$	34 000	75 500	3 450	7 700	6 500	9 500	**K42×47×27**	0.047
	48	17	25 700	46 000	2 630	4 700	6 500	9 500	**K42×48×17**	0.036
	50	20	34 000	53 500	3 450	5 500	6 500	9 500	**K42×50×20**	0.054
43	48	17	22 000	43 000	2 240	4 400	6 500	9 500	**K43×48×17**	0.029
	48	27 $^{-0.2}_{-0.55}$	34 000	75 500	3 450	7 700	6 500	9 500	**K43×48×27**	0.046
	50	18	29 100	49 000	2 960	5 000	6 500	9 500	**K43×50×18**	0.049
45	49	19	22 100	52 000	2 260	5 300	6 000	9 000	**K45×49×19**	0.027
	50	17	22 800	46 000	2 330	4 700	6 000	9 000	**KJ45×50×17S**	0.031
	50	27	34 500	78 000	3 500	7 950	6 000	9 000	**K45×50×27**	0.050
	52	18 $^{-0.2}_{-0.55}$	29 700	51 000	3 000	5 200	6 000	9 000	**K45×52×18**	0.051
	52	21 $^{-0.2}_{-0.55}$	32 000	56 500	3 300	5 750	6 000	9 000	**K45×52×21**	0.061
	53	20	36 000	59 000	3 650	6 000	6 000	9 000	**K45×53×20**	0.062
	53	25	46 500	82 000	4 700	8 400	6 000	9 000	**K45×53×25**	0.077
	53	28	49 500	90 000	5 050	9 200	6 000	9 000	**KJ45×53×28S**	0.078
47	52	17 $^{-0.2}_{-0.55}$	23 200	47 500	2 360	4 850	5 500	8 500	**K47×52×17**	0.033
	52	27 $^{-0.2}_{-0.55}$	35 500	83 000	3 650	8 450	5 500	8 500	**K47×52×27**	0.051
48	53	37.5 $^{-0.2}_{-0.55}$	41 500	101 000	4 200	10 300	5 500	8 500	**K48×53×37.5ZW**	0.072

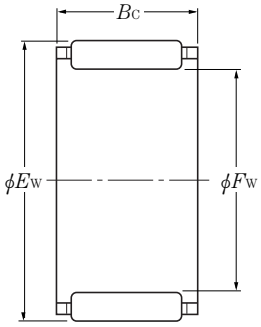
Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.



Type KMJ



Type KJ··S



Type KV··S

F_W 48～60mm

Boundary dimensions mm			Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Mass kg
F_w	E_w	B_c	C_r	C_{or}	C_r	C_{or}				(approx.)
48	54	19 $^{-0.2}_{-0.55}$	31 000	61 000	3 150	6 250	5 500	8 500	**K48×54×19**	0.044
	55	13.5	18 100	35 500	1 850	3 600	5 500	8 000	**K50×55×13.5**	0.023
	55	20	27 900	62 000	2 850	6 300	5 500	8 000	**K50×55×20**	0.042
	55	27	37 000	88 500	3 750	9 000	5 500	8 000	**K50×55×27**	0.053
50	55	30 $^{-0.2}_{-0.55}$	39 500	97 000	4 050	9 900	5 500	8 000	**K50×55×30**	0.059
	57	18	31 500	57 000	3 200	5 800	5 500	8 000	**K50×57×18**	0.053
	58	20	38 500	67 500	3 950	6 850	5 500	8 000	**K50×58×20**	0.065
	58	25	48 500	90 000	4 950	9 150	5 500	8 000	**K50×58×25**	0.081
52	58	19 $^{-0.2}_{-0.55}$	32 000	65 500	3 250	6 650	5 000	7 500	**K52×58×19**	0.048
	60	17	25 800	58 000	2 630	5 900	5 000	7 500	**K55×60×17**	0.043
	60	20	28 800	66 500	2 940	6 750	5 000	7 500	**K55×60×20**	0.045
	60	30	42 000	108 000	4 300	11 000	5 000	7 500	**KV55×60×30S**	0.069
	61	19	33 000	69 500	3 350	7 100	5 000	7 500	**K55×61×19**	0.051
55	61	20 $^{-0.2}_{-0.55}$	33 000	69 500	3 350	7 100	5 000	7 500	**K55×61×20**	0.054
	61	30	48 000	113 000	4 900	11 500	5 000	7 500	**K55×61×30**	0.081
	62	18	33 500	63 000	3 400	6 450	5 000	7 500	**K55×62×18**	0.054
	63	20	39 000	70 000	3 950	7 100	5 000	7 500	**K55×63×20**	0.073
	63	25	50 500	97 500	5 150	9 950	5 000	7 500	**K55×63×25**	0.088
	63	32	61 000	125 000	6 200	12 700	5 000	7 500	**K55×63×32**	0.117
58	64	19 $^{-0.2}_{-0.55}$	34 000	73 500	3 450	7 500	4 700	7 000	**K58×64×19**	0.052
	65	20	29 800	71 500	3 050	7 300	4 300	6 500	**K60×65×20**	0.051
	65	30	43 500	116 000	4 450	11 800	4 300	6 500	**K60×65×30**	0.071
	66	19	33 500	73 500	3 450	7 500	4 300	6 500	**K60×66×19**	0.053
	66	20	33 500	73 500	3 450	7 500	4 300	6 500	**K60×66×20**	0.056
60	66	30 $^{-0.2}_{-0.55}$	49 000	119 000	5 000	12 200	4 300	6 500	**K60×66×30**	0.084
	68	15	27 200	45 500	2 780	4 650	4 300	6 500	**K60×68×15**	0.058
	68	20	40 000	75 000	4 100	7 650	4 300	6 500	**K60×68×20**	0.077
	68	23	44 500	85 000	4 500	8 700	4 300	6 500	**K60×68×23**	0.092
	68	25	52 000	105 000	5 300	10 700	4 300	6 500	**K60×68×25**	0.097
	68	27	52 000	105 000	5 300	10 700	4 300	6 500	**K60×68×27**	0.098

Remarks: Even when an order is placed with reference to “Nominal bearing number” listed in this table, the ordered bearings are subject to being delivered with different cage type.